

REWARDING THE GRAPH BEHIND THE CHAIN: TOPOLOGY-AWARE PROCESS SUPERVISION FOR RL AND REASONING DISTILLATION

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ABSTRACT

Long-form reasoning is generated as a token sequence even when a conclusion depends on several non-adjacent premises. Final-answer rewards cannot expose this support structure, whereas learned process verifiers add supervision and inference cost. We introduce TopoPRM, an implicit process reward that deterministically constructs a typed support DAG from ordinary reasoning text. The same graph supplies support-completeness and local-continuity signals to an outcome-anchored hierarchical reward, an asymmetric advantage estimator (ACE) that prevents structural credit from reversing correctness, and topology-guided on-policy distillation from a fixed same-scale teacher. The resulting policy emits unmodified text and requires neither graph decoding nor a verifier at inference. On four mathematics benchmarks, the complete reward is the only reported variant that is non-dominated in both pass@1 and mean response length. Relative to outcome-only GRPO, it improves pass@1 by 0.2–13.3 points while reducing mean responses by 57–275 tokens. Full exceeds the no-topology checkpoint by 3.0 points on OlympiadBench and 10.0 points on AIME 2024.

1 INTRODUCTION

Reinforcement learning with verifiable rewards (RLVR) now drives long chain-of-thought (CoT) reasoning in large language models and is a core mechanism for training mathematical and symbolic ability (Wei et al., 2022; Kojima et al., 2022; DeepSeek-AI, 2025; Yeo et al., 2025; Shao et al., 2024; Ke et al., 2025). From sparse final-answer rewards (Cobbe et al., 2021) to process reward models (PRMs) that attach supervision to intermediate positions or prefixes (Lightman et al., 2023; Wang et al., 2024; Uesato et al., 2022), the dominant training interface still treats a trace as a *linear sequence of steps*. However, long reasoning is rarely linear: a later conclusion depends on several non-adjacent premises, an intermediate equation is reused downstream, and subgoals branch before merging into the answer (Besta et al., 2024; Zhong et al., 2026). **Closing the structural mismatch** is the core challenge for moving reasoning supervision beyond linear credit assignment.

Recent reasoning models sharpened the macroscopic structure of inference but stayed inside the CoT-as-result view. Outcome-only rewards scale (DeepSeek-AI, 2025; Yu et al., 2025; Lambert et al., 2024), yet they cannot separate a well-supported derivation from one that reaches the same answer through unsupported intermediate claims, and they remain open to reward hacking (Gao et al., 2023; Skalse et al., 2022). PRMs densify the signal with Monte Carlo rollouts, trained verifiers, or implicit outcome-derived scores (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2025; Khalifa et al., 2025; Zhao et al., 2026; Yuan et al., 2025; She et al., 2025), and can condition on a full prefix, but none explicitly recovers the **multi-parent support relation** in which one intermediate result supports, or is supported by, several non-adjacent steps (Wang et al., 2025). Token-level distillation likewise never requires a compact target to preserve those cross-step dependencies (Agarwal et al., 2024).

We propose TopoPRM, which supervises reasoning through its *implicit topological dependency structure* instead of assuming that surface order is already valid (Figure 1). A frozen, order-agnostic edge encoder recovers a raw directed support graph $\mathcal{H}_y$ from an unchanged free-form trace. Projecting $\mathcal{H}_y$ onto a text-order DAG $\mathcal{G}_y$ is what makes the discarded structure informative: backward-edge mass measures direction inconsistency, and cycle-edge mass measures circular support. We keep

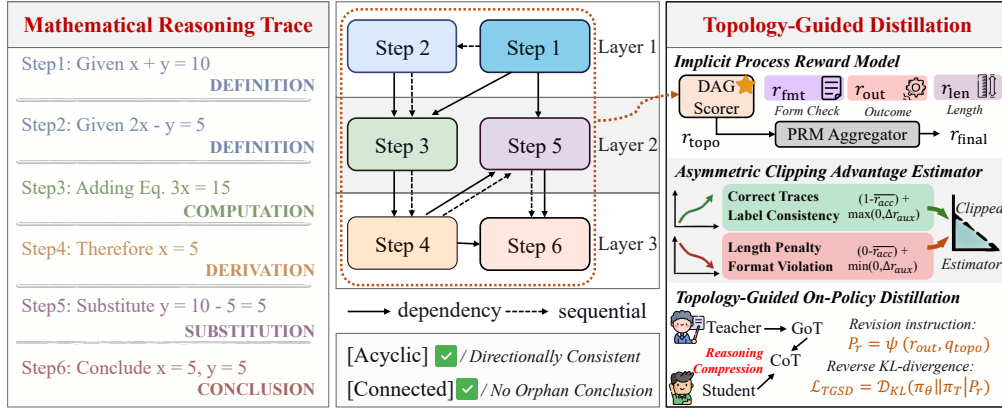


Figure 1: TopoPRM overview. The mathematical trace is mapped to a layered dependency graph; direction and acyclicity enter the topology-conditioned multi-source reward, while the same graph diagnostics localize dependency-preserving teacher revisions for distillation.

these as two separate reward sources and aggregate them non-additively with outcome, format, and continuity. Direction–acyclicity synergy then rewards traces that satisfy local support order and global non-circularity together, while the outcome and continuity interactions let topology reorder rollouts rather than act as a small generic bonus. The same projected graph guides distillation toward a compact dependency-preserving trace, and the policy never decodes graph tags or a special DAG format. Our contributions are threefold:

1. **Auditable structural supervision from free text.** We construct a typed support DAG without step labels, graph-formatted decoding, or a learned inference-time verifier, and separate typed support completeness from local continuity.
2. **A unified topology-guided post-training pipeline.** Hierarchical aggregation makes structure condition an outcome-dominant reward; ACE confines structural re-ranking to correctness strata; and TGD targets teacher supervision at unsupported or discontinuous regions.
3. **Accuracy–efficiency and audit evidence.** Across three post-training lineages, Full attains the highest displayed macro-average in every block. Controlled comparisons, training trajectories, edge audits, graph interventions, and cross-backbone stress tests characterize when the gains hold and where the surface extractor fails.

2 METHOD

TopoPRM organizes post-training around one frozen structural interface $\mathcal{E}$ (Figure 2). Stage I makes free-form traces consistently segmentable, Stage II uses $\mathcal{E}$ to constrain policy credit, and Stage III reuses the same evidence to request targeted revisions. This shared interface couples optimization and revision without making graph generation part of the policy.

2.1 TOPOLOGY-AWARE REASONING REPRESENTATION

For an input $x = (q, c)$, comprising a question q and optional context c , let π_θ generate a reasoning trace $y = (s_1, \dots, s_T)$. Outcome-only RLVR observes $r_{\text{out}}(x, y)$ but not the support relations among intermediate steps, and therefore assigns identical supervision to differently structured traces with the same outcome. TopoPRM augments this signal with a frozen interface $\mathcal{E}(y) = (\mathcal{G}_y, \mathcal{L}_y)$, where $\mathcal{G}_y$ is a surface-evidenced support DAG and $\mathcal{L}_y$ records the type and triggering evidence of each edge. The representation exposes recoverable cross-step support without treating it as semantic verification. It is used only during post-training: the policy emits ordinary text and does not invoke $\mathcal{E}$ at inference.

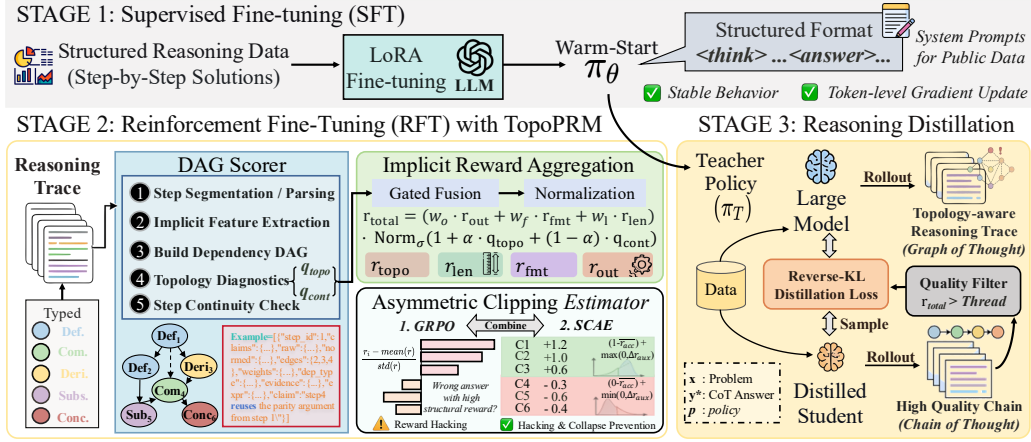


Figure 2: The TopoPRM pipeline. A frozen edge encoder converts free-form traces into typed support graphs. Stage II combines an outcome-anchored hierarchical reward with ACE; Stage III uses the same diagnostics to target on-policy teacher supervision. The Topology-guided Distillation (TGD) aims at improving reasoning behavior rather than compressing model size.

Surface-evidenced DAG construction. The extractor first segments a trace into ordered steps, then records normalized expressions, claim keys, variables, and coarse step types. Within each subproblem, it adds $i \rightarrow j$ for $i < j$ when expression reuse, claim reference, expression overlap, or variable reuse provides surface evidence. A derivation or conclusion without higher-confidence support may receive a lower-weight implicit-block edge from s_{j-1} ; eligible adjacent steps may also receive a sequential-fallback edge. Each ledger entry retains its endpoints, type, triggering evidence, and weight, making every score contribution traceable at the rule level (Appendix C).

Restricting candidates to $i < j$ makes $\mathcal{G}_y$ acyclic by construction. Direction and acyclicity are therefore extraction invariants rather than evidence of mathematical correctness. Model and manual judgments enter only the offline edge audit; every reported policy uses this frozen deterministic path.

Complementary structural diagnostics. Write $\mathcal{G}_y = (\mathcal{V}_y, \mathcal{A}_y)$ and let $\mathcal{C}_y$ denote its conclusion nodes. For edge weights ω_{ij} , define $\sigma_j = \min(1, \sum_{i:i \rightarrow j} \omega_{ij})$ and $\rho_{\text{orph}} = |\mathcal{C}_y|^{-1} \sum_{j \in \mathcal{C}_y} (1 - \sigma_j)$, taking $\rho_{\text{orph}} = 0$ when $\mathcal{C}_y$ is empty. Let $\delta(\mathcal{G}_y)$ be the fraction of order-consistent edges, and let $\kappa_{\text{edge}}(\mathcal{G}_y, \mathcal{G}^*)$ be support-edge F1 over endpoint pairs against an optional reference graph. We define

$$q_{\text{topo}}(y) = \frac{\lambda_b \mathbb{1}[\mathcal{V}_y] > 0 + \lambda_a \mathbb{1}[\mathcal{G}_y \text{ acyclic}] + \lambda_o \mathbb{1}[\rho_{\text{orph}} = 0] + \lambda_d \delta + \lambda_k \kappa_{\text{edge}}}{\lambda_b + \lambda_a + \lambda_o + \lambda_d + \lambda_k}, \quad (1)$$

where the weights are non-negative and have a positive sum; $\lambda_k = 0$ when no aligned reference is available. Policy optimization therefore uses only rollout graphs, since generated and reference steps are not aligned. Under forward-only extraction, direction and acyclicity are invariant contract guards, while graph existence and orphan conclusions provide the varying terms. The guards remain explicit so malformed or externally supplied graphs expose violations through the same interface.

Continuity provides the complementary local view. Set $h_i = 1$ when s_i contains an explicit problem-reference cue or reuses an expression or claim from a preceding step, and $h_i = 0$ otherwise. When neither cue is extractable, the scorer sets $h_i = 1$ to avoid conflating parser non-coverage with discontinuity. For $T > 0$, let $\eta(y) = T^{-1} \sum_{i=1}^T h_i$ and set $q_{\text{cont}} = 1$ when $\eta = 1$ and $q_{\text{cont}} = 0.8\eta$ otherwise; when $T = 0$, set $q_{\text{cont}} = 0$ (Appendix C). The varying orphan term in q_{topo} tests whether every identified conclusion accumulates sufficient incoming support, whereas q_{cont} measures stepwise traceability. Neither score verifies semantic correctness.

Why the two views are complementary. Neither view subsumes the other: a trace may end in an orphan conclusion despite fluent transitions, while non-local support need not ground every step. Each score resolves to parsed evidence, preserving this distinction without graph-formatted decoding or opaque process scores.

2.2 CORRECTNESS-CONSTRAINED STRUCTURAL OPTIMIZATION

Structural supervision must rank rollouts without overriding answer correctness. A free-standing bonus can offset an incorrect outcome, whereas a hard correctness gate discards useful variation among incorrect traces. We therefore separate reward ranking from admissible gradient updates.

Outcome-anchored hierarchical reward. The outcome-anchored base combines binary answer correctness $r_{\text{out}} \in \{0, 1\}$ with normalized format and length scores $r_{\text{fmt}}, r_{\text{len}} \in [0, 1]$ using non-negative weights that sum to one:

$$r_{\text{base}} = w_o r_{\text{out}} + w_f r_{\text{fmt}} + w_l r_{\text{len}}. \quad (2)$$

Within each GRPO group, each structural score is min-max rescaled to $[0, 1]$; a constant component maps to $1/2$. Here $\alpha_{\text{mix}} \in [0, 1]$ controls the topology-continuity mixture and $\epsilon > 0$ is the base floor. The hierarchical reward is

$$r_{\text{total}} = \text{clip}_{[0,1]}(\max(r_{\text{base}}, \epsilon) [1 + \alpha_{\text{mix}} \tilde{q}_{\text{topo}} + (1 - \alpha_{\text{mix}}) \tilde{q}_{\text{cont}}]). \quad (3)$$

The multiplicative form scales the structural adjustment by the outcome-anchored base rather than adding a base-independent bonus. Before clipping, the derivatives with respect to $(\tilde{q}_{\text{topo}}, \tilde{q}_{\text{cont}})$ are $\max(r_{\text{base}}, \epsilon)(\alpha_{\text{mix}}, 1 - \alpha_{\text{mix}})$. Because clipping is monotone, it preserves the weak order of the unclipped scalar rewards but may create ties; the estimator below additionally constrains the sign of each policy-gradient coefficient.

Asymmetric credit estimation (ACE). Following correctness-stratified clipping for graph-formatted reasoning (Liu et al., 2026), ACE applies the same admissibility principle to structure extracted from free text. Let z_i be the group-standardized value of $r_{\text{total}}^{(i)}$ and Δu_i the correctness-stratum-centered topology-continuity score; Appendix A.3 specifies both, including the zero-variance convention. Let $\mathcal{C} = \{i : r_{\text{out}}^{(i)} = 1\}$ and $\mathcal{W} = \{i : r_{\text{out}}^{(i)} = 0\}$ denote correct and incorrect rollouts, and let $\underline{c} \leq 0 \leq \bar{c}$ be the lower and upper coefficient bounds. ACE computes

$$\hat{A}_i = \begin{cases} \text{clip}(z_i + \max(0, \Delta u_i), 0, \bar{c}), & i \in \mathcal{C}, \\ \text{clip}(z_i + \min(0, \Delta u_i), \underline{c}, 0), & i \in \mathcal{W}. \end{cases} \quad (4)$$

Stage I supplies a parsable supervised warm start. From this initialization, Stage II optimizes Eq. 3 with GRPO (Shao et al., 2024) and realizes its updates through Eq. 4.

Proposition 1 (Sign separation under one-sided structural adjustment). *Assume $\underline{c} \leq 0 \leq \bar{c}$. Then $\hat{A}_i \in [0, \bar{c}]$ for $i \in \mathcal{C}$ and $\hat{A}_i \in [\underline{c}, 0]$ for $i \in \mathcal{W}$. Before saturation, a positive structural deviation can only increase the coefficient of a correct rollout, while a negative deviation can only decrease that of an incorrect rollout. The branchwise clips therefore enforce non-negative coefficients on $\mathcal{C}$ and non-positive coefficients on $\mathcal{W}$, although saturation may create ties.*

Equation 3 assigns bounded rollout rewards, whereas Proposition 1 constrains the corresponding advantage coefficients. Appendix A.3 provides the derivation. The *w/o ACE* control retains the structural reward and final distillation stage, providing the closest estimator comparison within the complete pipeline.

2.3 DEFECT-CONDITIONED TOPOLOGY-GUIDED DISTILLATION (TGD)

Stage III pairs the Stage-II target π_θ with a fixed teacher π_T of matching architecture and parameter count. From an on-policy trace y^* , the scorer selects a defect localized by the surface-evidence graph—an orphan conclusion, weak non-local support, or a continuity break—and encodes the triggering evidence in revision prompt P_r . The teacher conditions on (x, y^*, P_r) , and the indicator $a_r \in \{0, 1\}$ accepts a revision only if it has a correct final answer and valid format, does not decrease either structural diagnostic, and remains within the prescribed response budget.

Gated distillation objective. On a teacher-revised prefix $\tilde{y}_{<t}$, let $p_t = \pi_\theta(\cdot \mid x, \tilde{y}_{<t})$ and $q_t = \pi_T(\cdot \mid x, y^*, P_r, \tilde{y}_{<t})$. The target minimizes the token-summed reverse-KL objective

$$\mathcal{L}_{\text{TGD}}(\theta) = \mathbb{E}_{x, y^* \sim \pi_\theta(\cdot \mid x), \tilde{y} \sim \pi_T(\cdot \mid x, y^*, P_r)} a_r(x, y^*, \tilde{y}) \sum_{t=1}^{|\tilde{y}|} \mathcal{D}_{\text{RKL}}(p_t \parallel q_t). \quad (5)$$

Stage III transfers behavior between equal-size checkpoints rather than compressing the model; every complete TopoPRM checkpoint includes this stage. The graph specifies where revision should act rather than what the model should output; accepted targets remain ordinary reasoning text. After segmentation, deterministic extraction enumerates $O(T^2)$ candidate step pairs, invokes no language model, and is removed at inference.

3 EXPERIMENTS

The evaluation addresses four questions. RQ1 asks whether gains recur across model lineages and where Full lies on the accuracy–length frontier. RQ2 examines sensitivity to credit constraints, topology, and continuity. RQ3 tests extracted support against separately produced references, and RQ4 asks whether topology adds ranking information beyond a semantic PRM. Appendix D provides repeated-decoding, structure–correctness, cross-backbone, and shared-pool tests.

Table 1: Pass@5 accuracy (%) on nine public benchmarks. The upper block provides complete open-checkpoint references; the lower blocks compare post-training methods within the same model family. **Bold** and underlined values denote the best and second-best result within each post-training block, respectively; tied best results are bold. Avg. is the reported nine-task macro-average. All values are preserved from the archived five-sample result ledger; Appendix B separates this protocol from the controlled pass@1 study and records the provenance scope.

Model	Params	Elementary		Olympiad		Competition			General		Avg.
		GSM8K	MATH	Olympiad	Omni	AIME'24	AIME'25	CNMO'24	MMLU	GPQA-D	
Complete open checkpoints											
Qwen3.5-2B	2B	77.2	60.8	48.5	13.3	6.7	13.3	36.1	32.8	37.9	36.3
Qwen3.5-4B	4B	85.1	55.8	32.8	43.4	0.0	13.3	10.0	67.4	29.8	37.5
Qwen2.5-Math-7B-Instruct	7B	95.2	73.8	38.0	22.0	13.0	10.0	13.3	54.8	28.3	38.7
Llama-3.1-8B-Instruct	8B	84.5	52.2	15.0	10.0	3.0	2.0	3.0	68.0	30.0	29.7
DeepSeek-R1-Distill-Llama-8B	8B	83.0	89.1	48.0	25.0	50.4	35.0	25.0	52.0	49.0	50.7
Qwen2.5-7B post-training											
Base	7B	85.4	49.8	25.0	18.0	7.0	5.0	8.0	74.2	36.4	34.3
+ SFT	7B	91.3	65.4	42.5	54.2	16.7	12.2	16.7	70.1	28.3	44.2
+ SimpleRL-Zoo	7B	91.7	78.2	38.6	40.4	20.0	15.6	15.0	62.5	55.2	46.4
+ Open-Reasoner-Zero	7B	89.0	81.4	47.9	38.4	17.9	15.6	13.3	62.5	36.6	44.6
+ TopoPRM (Ours)	7B	95.8	72.6	47.8	57.2	16.7	26.7	47.0	82.7	51.0	55.3
DeepSeek-R1-Distill-Qwen-7B post-training											
Base	7B	73.8	36.8	25.8	28.0	46.7	30.0	22.9	23.7	12.6	33.4
+ SFT	7B	86.0	92.8	52.0	48.0	55.5	40.0	30.0	50.0	49.1	55.9
+ GRPO	7B	85.1	67.4	56.2	72.0	50.0	40.0	57.8	50.1	17.7	55.1
+ TopoPRM (Ours)	7B	84.3	66.6	56.4	72.2	55.5	40.0	55.4	60.3	34.3	58.3
Qwen3.5-9B post-training											
Base	9B	69.5	66.6	55.2	61.5	16.7	21.1	16.7	28.6	37.4	41.5
+ SFT	9B	94.1	68.2	44.0	53.8	30.0	15.6	30.0	78.4	47.0	51.2
+ GRPO	9B	93.3	63.0	56.2	72.0	46.7	30.0	57.8	63.0	17.7	55.5
+ TopoPRM (Ours)	9B	97.9	81.9	56.2	76.2	43.3	18.9	47.0	86.4	40.4	60.9

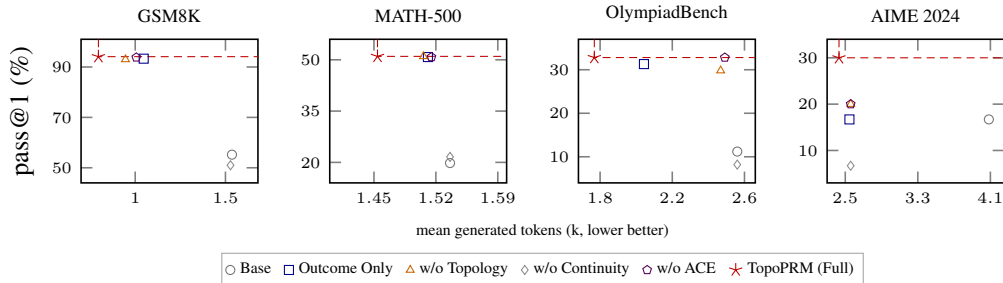


Figure 3: Accuracy–length trade-off among the Qwen3.5-9B checkpoints in Table 2. Full (★) is the only checkpoint on the observed Pareto frontier of all four benchmarks. Axes are cropped per panel; length is in thousands of visible tokens.

Table 2: End-to-end ablation on Qwen3.5-9B from a shared Stage-I initialization. Cells report pass@1 (%) / mean visible tokens; the last columns give the four-task macro-average and mean length. Gold/blue shading marks the TGD-matched control/Full.

Configuration	TGD	pass@1 ↑ / tokens ↓				Avg. pass@1 ↑	Mean tokens ↓
		GSM8K	MATH-500	OlympiadBench	AIME 2024		
<i>Stage-I reference</i>							
Base	–	55.2 / 1,536	19.8 / 1,536	11.2 / 2,560	16.7 / 4,083	25.7	2,429
<i>Post-training variants</i>							
Outcome Only	No	93.3 / 1,047	50.8 / 1,511	31.3 / 2,043	16.7 / 2,546	48.0	1,787
w/o Topology	No	93.0 / 946	51.0 / 1,506	29.8 / 2,468	20.0 / 2,560	48.5	1,870
w/o Continuity	No	51.0 / 1,528	21.6 / 1,536	8.2 / 2,560	6.7 / 2,560	21.9	2,046
w/o ACE	Yes	93.8 / 1,006	50.8 / 1,515	32.8 / 2,491	20.0 / 2,560	49.4	1,893
TopoPRM (Full)	Yes	94.1 / 796	51.0 / 1,454	32.8 / 1,768	30.0 / 2,431	52.0	1,612



Figure 4: Training and efficiency diagnostics from a complementary run suite. Trajectories summarize reward, response length, and Acc/kTok; bars and diamonds show endpoint length and efficiency. Objective scales differ, and these panels are not pooled with the controlled comparison in Table 2.

3.1 EVALUATION SETUP

Training, baselines, and controls. All controlled variants share a Qwen3.5-9B Stage-I checkpoint (Qwen Team, 2026a) and frozen graph extractor. Stage II applies rank-64 LoRA (Hu et al., 2022) with GRPO (Shao et al., 2024), two completions per prompt, and learning rate 5×10^{-6} . Full and w/o ACE proceed to Stage III with a same-architecture, equal-size teacher; *Base* is Stage I, and other controls stop after Stage II.

Table 1 groups complete Qwen3.5, Qwen2.5-Math, Llama 3.1, and DeepSeek-R1-Distill checkpoints (Qwen Team, 2026b;c; Yang et al., 2024b; Grattafiori et al., 2024; DeepSeek-AI, 2025) with within-lineage SFT, GRPO, SimpleRL-Zoo (Zeng et al., 2025), and Open-Reasoner-Zero (Hu et al., 2025) baselines. Because external data and compute differ, Table 2 supplies the shared-initialization analysis. *Outcome Only* uses final-answer correctness, and *w/o ACE* retains the structural reward but uses standard group-relative advantages. The topology and continuity controls remove the named source and redistribute its reward mass. The shorter Stage-II training cap for *w/o Continuity* makes the source-removal rows tests of pipeline sensitivity rather than factorial ablations (Appendix A).

Benchmarks and metrics. Table 1 reports five-sample pass@5 across its nine benchmark columns (Cobbe et al., 2021; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024a; Hendrycks et al., 2020; Rein et al., 2023). The controlled study reports pass@1 and mean visible tokens on GSM8K, MATH-500 (Lightman et al., 2023), a 134-item English text-only OlympiadBench subset,

and AIME 2024. Acc/kTok is $\text{pass}@1/(\sqrt{\text{tokens}}/1000)$ and is not optimized. Figure 4 uses a separate suite; Appendix A records all three protocols.

Each result source is interpreted only within its own protocol: Table 1 measures breadth, Table 2 controls initialization and response cost, and Figure 4 summarizes training behavior.

3.2 MAIN RESULTS: ACCURACY AND EFFICIENCY (RQ1)

Observation 1: The complete configuration leads every evaluated post-training lineage in aggregate. TopoPRM reaches macro-averages of 55.3, 58.3, and 60.9 on Qwen2.5-7B, DeepSeek-R1-Distill-Qwen-7B, and Qwen3.5-9B, exceeding the strongest alternative in each block by 8.9, 2.4, and 5.4 points (Table 1). It is best or tied-best on five of nine tasks in every lineage, although the winning tasks vary by backbone. Each TopoPRM row records a best or tied-best result in at least three of the four task categories; the Qwen2.5-7B variant spans all four. The Qwen3.5-9B average of 60.9 is also the table maximum, 10.2 points above the strongest complete-checkpoint reference. This profile is not uniformly dominant: Qwen3.5 GRPO remains stronger on several competition sets, whereas TopoPRM gains most on MATH, Omni-MATH, MMLU, and GPQA-D. The evidence therefore supports broad aggregate improvement under the reported protocol, not universal task-wise dominance.

For scale context, Qwen2.5-7B TopoPRM (55.3) exceeds the larger DeepSeek-R1-Distill-Llama-8B reference (50.7). Because their data and compute differ, this is contextual rather than a controlled scaling effect; component attribution relies on shared initialization.

Observation 2: Full is the only checkpoint on each observed accuracy-length frontier. Against Outcome Only, Full improves $\text{pass}@1$ by 0.8, 0.2, 1.5, and 13.3 points on GSM8K, MATH-500, OlympiadBench, and AIME 2024 while saving 251, 57, 275, and 115 tokens (Figure 3; Table 2). The easier tasks primarily separate response cost, whereas AIME supplies the largest accuracy gap. Acc/kTok rises from 89.1 to 118.2 on GSM8K and from 6.6 to 12.3 on AIME, with the same direction elsewhere. Full is therefore the only checkpoint on all four observed frontiers. Each variant is trained once, so this comparison does not estimate training-seed uncertainty.

The two evaluations rule out complementary explanations. Table 1 shows that the aggregate gain recurs across post-training lineages but does not control response cost, whereas Table 2 shows that Full improves or preserves accuracy while shortening responses on every benchmark. The gain is therefore not explained by greater verbosity, and the length reduction is not merely truncation: the largest accuracy increase occurs on AIME, where premature stopping would be most damaging.

3.3 ABLATION STUDY (RQ2)

Observation 3: The distillation-matched checkpoint comparison favors Full; source removals expose pipeline sensitivity. Full versus *w/o ACE* is the only contrast matched through distillation. Full raises four-task average $\text{pass}@1$ from 49.4 to 52.0 and reduces mean length from 1,893 to 1,612 tokens (Table 2). On the three larger sets, accuracy is nearly unchanged while responses are 61–723 tokens shorter; on AIME, the 10.0-point gap equals three problems. Removing topology costs 3.0 and 10.0 points on OlympiadBench and AIME 2024, respectively, while adding 700 and 129 tokens. Removing continuity costs 23.3–43.1 points and lengthens every task, consistent with local traceability and non-local support acting jointly. Because source removals stop before distillation and the continuity cap differs, they measure pipeline sensitivity rather than isolated effects. Post-extraction controls test diagnostic selectivity separately: reversal changes the direction channel by -0.715 , whereas matched cycle closure changes only acyclicity, by -0.328 (Appendix C.3). These are graph interventions, not trained-policy effects.

3.4 AUDITING RECOVERABLE SUPPORT (RQ3)

Audit protocol. Traces are stratified by task and response length. Both model annotators receive identical parsed pairs but neither extractor decisions nor edge types. After independent labeling, the complete set undergoes manual error checking. We report agreement and extractor scores against each model separately, preserving the distinction between annotation stability and extractor-reference agreement without treating the checks as a human gold standard.

Table 3: Held-out audit on 1,770 candidate pairs from 119 traces. Qwen3-32B and Qwen2.5-32B label pairs independently; manual verification covers the complete set. Reported extractor metrics use the model annotations as references. Consensus/union are their positive-label intersection/union.

Annotator contingency Qwen3-32B vs. Qwen2.5-32B				Released extractor against alternative references			
Qwen3-32B	Support	No support	Total	Support reference	Prec.	Recall	F1
Support	368	68	436	Qwen3-32B	.478	.589	.528
No support	113	1,221	1,334	Qwen2.5-32B	.483	.541	.510
Total	481	1,289	1,770	Annotator consensus	.428	.625	.508
Agreement: raw = 0.898; κ = 0.734				Annotator union	.534	.523	.528

Observation 4: Separately produced references are consistent, while extractor–reference agreement remains moderate. Qwen3-32B and Qwen2.5-32B independently labeled 1,770 candidate pairs from 119 traces. They agreed on 89.8%, with Cohen’s $\kappa_{\text{ann}} = 0.734$ (Table 3); their positive-label rates are 24.6% and 27.2%, with 368 jointly positive and 181 disputed pairs. Every pair was also checked for evident annotation or parsing errors. Extractor F1 is 0.528 against Qwen3-32B and 0.510 against Qwen2.5-32B; consensus and union references yield 0.508 and 0.528. This narrow range indicates limited sensitivity to the chosen Qwen-family reference, while the moderate absolute F1 precludes a proof-level interpretation. The 120-trace typed ledger attributes 145 false positives to variable-reference edges, compared with 34 expression-reference, 11 expression-overlap, 9 implicit-block, and 82 order-fallback edges. A guarded rule revision raises precision from 0.480 to 0.580 on that ledger. Because the same labels informed its design and evaluation, the result is post-hoc rather than a held-out improvement (Appendix C.5).

3.5 BOUNDARY ANALYSIS WITH A SEMANTIC PRM (RQ4)

Table 4: Diagnostic PRM–topology fusion on shared Best-of-8 pools ($n = 80$ per benchmark). The hybrid multiplies normalized Qwen2.5-Math-PRM-7B scores by $1 + q_{\text{topo}}$; Δ is the change from PRM selection.

Benchmark	PRM@8	PRM×Topo@8	Δ	Oracle@8
GSM8K	95.00	95.00	0.00	96.25
MATH-500	83.75	83.75	0.00	92.50

Observation 5: Topology does not replace semantic scoring. On shared Best-of-8 pools, multiplying Qwen2.5-Math-PRM-7B scores (Zhang et al., 2025) by $1 + q_{\text{topo}}$ ties the PRM on GSM8K (95.00%) and MATH-500 (83.75%; Table 4). Topology-only selection trails the PRM on both tasks, while the pool oracle remains 1.25 and 8.75 points higher. The tie bounds the claim: this fusion neither improves aggregate selection nor replaces semantic verification. Establishing complementarity requires larger prespecified pools and item-level disagreement analysis.

3.6 ROBUSTNESS AND FAILURE MODES

Observation 6: Cross-backbone accuracy is mixed, and stochastic decoding does not preserve the greedy ordering. In the matched DeepSeek-R1-Distill-Qwen-7B study, greedy pass@1 favors Full, but five-draw means span only 67.8–68.0 with standard deviations of 1.8–2.7, so the ordering is not stable under decoding variation. On Llama-3.1-8B-Instruct, Full saves 389 tokens on GSM8K with unchanged pass@1 and 396 tokens on MATH-500 while pass@1 falls from 46.5 to 42.5. Response shortening transfers to these subsets; accuracy ordering and preservation do not (Appendix D).

Observation 7: Topology score is associated with correctness but is not uniquely discriminative. Full has the highest mean topology score (0.939), and its high-score stratum exceeds its low-score stratum by 17.15 accuracy points. Outcome-only training exhibits a comparable 15.04-point gap. The PRM tie and edge-audit F1 likewise support interpreting graph diagnostics as an inspectable structural preference under outcome verification, not as a semantic discriminator.

3.7 EVIDENCE SYNTHESIS AND SCOPE

Taken together, the results support a bounded but coherent claim: Full leads the reported aggregates and occupies the observed Qwen3.5-9B accuracy–length frontiers without purchasing accuracy through longer responses. Pipeline comparisons and graph interventions reveal sensitivity to credit constraints, non-local support, and continuity, while the edge audit keeps the resulting signal inspectable despite moderate extractor–reference agreement. The PRM and cross-backbone analyses further delimit this interpretation: topology provides a training-time structural preference rather than semantic verification or a universally transferable accuracy signal. Training-seed stability and broader backbone generalization remain open (Appendix E).

4 RELATED WORK

Reward supervision for reasoning. RLHF learns preference rewards (Ouyang et al., 2022), whereas RLVR uses programmatic answer checks (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024; Zeng et al., 2025); subsequent objectives improve optimization and credit propagation (Yu et al., 2025; Kazemnejad et al., 2025). Terminal rewards, however, cannot distinguish differently supported traces with the same answer and remain vulnerable to gaming (Gao et al., 2023; Skalse et al., 2022; Gao et al., 2024b). PRMs provide denser feedback through human labels (Uesato et al., 2022; Lightman et al., 2023), rollout or search (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2024), induced annotations (Lu et al., 2024), outcome-derived values (Yuan et al., 2025; Cui et al., 2026), or generative verification (Khalifa et al., 2025; Zhao et al., 2026); PRMBench and ProcessBench nevertheless reveal persistent error-detection and generalization failures (Song et al., 2025; Zheng et al., 2024). Scalar completion, step, or prefix scores summarize reasoning quality without explicitly identifying the non-adjacent premises supporting a conclusion. *TopoPRM complements answer verification with auditable local-continuity and non-local-support signals extracted from free text.*

Structured reasoning representations. Tree of Thoughts and Graph of Thoughts expose branching search and dependencies (Yao et al., 2023; Besta et al., 2024); PARC represents explicit premise links (Mukherjee et al., 2025), while DAG-Math and GraphPRM operate on graph-formatted derivations or trajectories (Zhang et al., 2026; Peng et al., 2025). Correctness-stratified clipping has also been studied for graph-formatted reasoning (Liu et al., 2026). *TopoPRM instead extracts evidence-bearing support from ordinary reasoning text and uses it for training-time credit assignment and targeted revision, without changing the policy output or inference interface.*

Reasoning distillation and compression. Knowledge distillation transfers teacher behavior (Hinton et al., 2015); language-model variants address exposure bias through reverse-KL, on-policy, adaptive off-policy, Best-of- N , black-box, or reward-coupled objectives (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024; Sessa et al., 2024; Xu et al., 2025; Ye et al., 2025; Yang et al., 2026). Reasoning compression controls budgets, prunes tokens, tunes for length, or shapes efficiency rewards (Snell et al., 2024; Muennighoff et al., 2025; Xia et al., 2025; Luo et al., 2025; Arora & Zanette, 2025; Aggarwal & Welleck, 2025); structured variants prune reasoning DAGs or distill semantic operations (Yuan et al., 2026; Wen et al., 2026). *TopoPRM uses defects detected in the student’s own trace to target teacher revision, while answer, structure, and response budget jointly determine which revisions become training targets.*

5 CONCLUSION

We presented TopoPRM, which infers evidence-backed dependencies from free-form reasoning and uses their topology to guide reinforcement learning and targeted distillation, addressing the structural blindness of outcome-only supervision. Because this supervision is confined to post-training and preserves ordinary text generation at inference, the framework can scale across model families and reasoning tasks without modifying the deployed architecture. In future work, we will improve the fidelity and semantic grounding of dependency inference and extend topology-aware supervision to broader reasoning domains and larger-scale training.

AI USE STATEMENT

Generative AI tools assisted with literature discovery, manuscript organization, and language polishing. The authors inspected the primary sources, code paths, experimental artifacts, equations, citations, and every AI-assisted edit. The authors take responsibility for the final text, claims, code, and results.

ETHICS STATEMENT

The reported study uses public mathematics benchmarks and model-generated reasoning traces. It collects no personal or demographic data, includes no human-subject study, and uses no private benchmark to support a claim. The edge audit uses model annotators and is identified as such. The principal risk is methodological: a trace can exhibit coherent surface dependencies while containing an incorrect mathematical deduction. We therefore treat topology as a training-time process signal rather than a semantic verifier, report this failure mode directly, and retain final-answer verification as a separate reward source. The study is limited to mathematical reasoning; broader deployment claims would require domain-specific safety and validity evaluation.

REPRODUCIBILITY STATEMENT

Appendix A records the training, reward, and decoding configuration, and Appendix C specifies graph construction and the audit protocol. The anonymized supplementary artifact contains source and configuration snapshots plus aggregate JSON files for the edge audit, controlled rerun, cross-backbone, structural, and shared-pool diagnostics. Table 1 was transcribed from archived aggregate evaluation tables; its per-item outputs are not included, which we disclose as a reproducibility limitation. We distinguish single training runs, repeated decoding draws, and paired annotation statistics throughout, and we do not use incomplete experiments as evidence.

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A IMPLEMENTATION AND EVALUATION DETAILS

A.1 TRAINING CONFIGURATION

Table 5 documents the shared SFT and GRPO configuration for the controlled Qwen3.5-9B suite. All variants use the same Stage-I initialization and frozen edge encoder. Full and *w/o ACE* additionally use the same TGD procedure and fixed same-parameter teacher checkpoint. Source-removal controls redistribute reward mass, and *w/o Continuity* uses a shorter completion cap; these rows are matched at initialization but are not factorial.

Table 5: Training configuration for the controlled Qwen3.5-9B runs.

Parameter	Stage I: SFT	Stage II: GRPO
Initialization	Qwen3.5-9B	Stage-I checkpoint
Adapter	LoRA $r = 64$, $\alpha = 128$	LoRA $r = 64$, $\alpha = 128$
Learning rate	5×10^{-5}	5×10^{-6}
Schedule / warmup	cosine / 3%	cosine / 5%
Training length	2 epochs	1 epoch
Microbatch / accumulation	2 / 8	1 / 16
Completions / prompt	–	2
Sampling / nucleus	–	0.8 / 0.95
Completion cap	–	4,096; 1,024 for w/o continuity
Maximum sequence	4,096	8,192; 4,096 for w/o continuity
Precision	bf16	bf16

A.2 REWARD CONFIGURATION

The released topology score uses $(\lambda_b, \lambda_a, \lambda_o, \lambda_d, \lambda_k) = (0.20, 0.15, 0.15, 0.15, 0.25)$; without a reference graph, λ_k is omitted and the remaining weights are renormalized. The hierarchical base uses $(w_o, w_f, w_l) = (0.70, 0.15, 0.15)$, $\alpha = 0.60$, and $\epsilon = 0.05$. Outcome-only uses r_{out} . The no-topology and no-continuity controls assign $(0.55, 0.15, 0.20, 0.10)$ to outcome, format, the retained structural term, and length. Thus, a removed source is replaced rather than simply set to zero in Eq. 3. The length score is one through 2,000 decoded characters, decreases linearly, and reaches zero at 4,000; all benchmarks use the same constants.

A structural component that is constant within a rollout group is mapped to $1/2$ before Eq. 3. Gaussian variance injection and dynamic reweighting are disabled. The supplementary snapshot contains the exact reward classes and defaults.

A.3 ACE AND TGD

ACE partitions each rollout group by final-answer correctness before centering the auxiliary structural contribution. Equation 4 clips the correct stratum to non-negative advantages and the incorrect stratum to non-positive advantages. When a stratum is absent, it contributes no update; a singleton stratum has zero centered auxiliary deviation. The *w/o ACE* row replaces this estimator with standard group-relative advantages while leaving TGD unchanged.

Proof of Proposition 1. The range claims follow directly from the two clipping intervals. For $i \in \mathcal{C}$, the admitted auxiliary term is $\max(0, \Delta r_{\text{aux}}^{(i)})$; hence it is monotone in a positive deviation and constant over non-positive deviations until clipping saturates. For $i \in \mathcal{W}$, $\min(0, \Delta r_{\text{aux}}^{(i)})$ is monotone over negative deviations and constant over non-negative deviations. If two rollouts share a stratum and standardized return, subtracting their unsaturated advantages cancels that return and leaves the difference between their admitted auxiliary terms. This proves sign separation, one-sided sensitivity, and within-stratum order preservation.

TGD evaluates the target policy on-policy, selects a graph-localized defect, and asks the fixed teacher for a revision distribution. Accepted revisions must satisfy answer and format checks, improve or preserve the topology and continuity diagnostics, and meet the response budget. Reverse-KL is evaluated on target-policy prefixes. Teacher and target have the same architecture and parameter count but are distinct checkpoints with fixed roles.

A.4 EVALUATION PROTOCOL

Public mathematics tasks use boxed-answer extraction and symbolic equivalence checking, without a rubric grader.

Pass@5 suite. For Table 1, pass@5 is $N^{-1} \sum_{i=1}^N \mathbb{1}[\max_{j \leq 5} c_{ij} = 1]$ over five samples at temperature 0.7 and top- $p = 0.95$. Completion caps are 1,536 tokens for GSM8K and MATH, 2,560 for competition and olympiad sets, and 512 for MMLU and GPQA-D. External training budgets are not matched; these rows contextualize end-to-end checkpoints rather than isolate a training treatment.

Controlled pass@1 suite. The controlled study decodes once per item and retokenizes visible text without special tokens. Table 14 instead varies decoding seeds for fixed policies. The former compares observed checkpoints; the latter measures decoding variation, not training-seed uncertainty. Neither protocol is pooled with the pass@5 ledger.

Optimization-log suite. Figure 4 retains online-batch Stage-II trajectories and endpoint aggregates on GSM8K, MATH-500, AIME 2024, and Omni-MATH. Its task mixture and aggregation follow the optimization logs and are not pooled with either primary table. The extended p@5/maj@5 cells below likewise retain their contemporaneous controlled-suite subsets and harness.

B EXTENDED RESULTS

Table 1 reports the nine-task pass@5 comparison at its recorded precision. Appendix tables below expose the checkpoint-level pass@1 and per-benchmark ablation measurements used for the accuracy-length analysis. Accuracy-length claims use only shared-initialization controls.

B.1 CONTROLLED PASS@1 CHECKPOINTS

The lower block supplies Figure 3 and Table 2; its variants share the Stage-I initialization. Full and w/o ACE include TGD, while the other controlled variants stop after Stage II. The upper block reports complete open checkpoints, with † marking checkpoints evaluated using our prompts and answer extractor. Accuracy-length claims use only the controlled lower block.

B.2 COMPLETE PER-BENCHMARK ABLATION LEDGER

Tables 7–10 report the complete eight-benchmark ablation ledger. The w/o ACE checkpoint retains TGD and is the closest control for the estimator; topology and continuity removals stop after Stage II and measure end-to-end pipeline sensitivity.

Table 6: Controlled pass@1 checkpoint comparison on four mathematics benchmarks. The upper block contains complete open checkpoints available in the supplied comparison sheet; † marks checkpoints evaluated with our prompts and answer extractor. The lower block contains the archived Qwen3.5-9B reward variants used for the accuracy-length analysis. Avg. is the macro-average over four tasks.

Method	Params	GSM8K	MATH-500	Olympiad Bench	AIME 2024	Avg.
<i>Complete open checkpoints</i>						
Qwen3.5-4B† (Qwen Team, 2026c)	4B	57.7	38.6	19.4	3.3	29.8
Qwen3.5-9B-Instruct† (Qwen Team, 2026a)	9B	66.6	41.2	26.1	3.3	34.3
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	84.5	52.2	15.0	3.0	38.7
Qwen2.5-7B-Instruct (Yang et al., 2024a)	7B	91.6	75.5	25.0	7.0	49.8
<i>Controlled Qwen3.5-9B post-training variants</i>						
Stage-I Base	9B	55.2	19.8	11.2	16.7	25.7
Outcome Only	9B	93.3	50.8	31.3	16.7	48.0
w/o Topology	9B	93.0	51.0	29.8	20.0	48.5
w/o Continuity	9B	51.0	21.6	8.2	6.7	21.9
TopoPRM (Full)	9B	94.1	51.0	32.8	30.0	52.0

Table 7: Detailed Qwen3.5-9B ablation ledger for GSM8K and MATH-500. Tok. is mean visible response length; Time is total evaluation wall-clock time in seconds.

Variant	GSM8K ($n = 1,319$)						MATH-500 ($n = 500$)					
	p@1	p@5	maj@5	Tok.	Time	Acc/kTok	p@1	p@5	maj@5	Tok.	Time	Acc/kTok
Base	55.2	87.8	55.7	1,536	45.0	35.9	19.8	39.2	21.4	1,536	18.7	12.9
Outcome Only	93.3	97.7	93.6	1,047	34.2	89.1	50.8	59.0	55.2	1,511	18.9	33.6
w/o Topology	93.0	97.7	93.9	946	25.0	98.3	51.0	59.4	55.6	1,506	16.2	33.9
w/o Continuity	51.0	86.1	52.6	1,528	48.3	33.4	21.6	39.6	23.2	1,536	19.1	14.1
ACE-off (archived)	93.8	98.0	94.6	1,006	28.3	93.2	50.8	59.8	55.2	1,515	18.2	33.5
TopoPRM (Full)	94.1	97.9	94.8	796	19.7	118.2	51.0	81.9	54.6	1,454	16.6	35.1

C GRAPH CONSTRUCTION AND EDGE AUDIT

C.1 DETERMINISTIC CONSTRUCTION

Algorithm 1 summarizes the released deterministic rule path. Optional language-model edge refinement is disabled in all reported policy runs.

Table 8: Detailed Qwen3.5-9B ablation ledger for OlympiadBench and Omni-MATH. Metric definitions follow Table 7.

Variant	OlympiadBench ($n = 134$)						Omni-MATH ($n = 262$)					
	p@1	p@5	maj@5	Tok.	Time	Acc/kTok	p@1	p@5	maj@5	Tok.	Time	Acc/kTok
Base	11.2	25.4	14.2	2,560	26.5	4.4	16.4	33.2	17.9	2,559	52.0	6.4
Outcome Only	31.3	46.3	37.3	2,043	19.7	15.3	41.6	55.3	48.9	1,845	35.2	22.5
w/o Topology	29.8	50.7	38.1	2,468	11.8	12.1	42.4	54.2	45.8	2,430	17.8	17.4
w/o Continuity	8.2	25.4	9.0	2,560	8.7	3.2	14.9	30.5	17.6	2,560	18.5	5.8
ACE-off (archived)	32.8	47.0	35.8	2,491	9.6	13.2	42.0	53.8	47.3	2,463	19.4	17.1
TopoPRM (Full)	32.8	50.7	38.8	1,768	18.1	18.6	42.8	55.7	47.7	1,592	19.0	26.9

Table 9: Detailed Qwen3.5-9B ablation ledger for AIME 2024 and AIME 2025. Metric definitions follow Table 7.

Variant	AIME 2024 ($n = 30$)						AIME 2025 ($n = 90$)					
	p@1	p@5	maj@5	Tok.	Time	Acc/kTok	p@1	p@5	maj@5	Tok.	Time	Acc/kTok
Base	16.7	33.3	33.3	4,083	8.5	4.1	2.2	4.4	1.1	2,560	5.9	0.9
Outcome Only	16.7	33.3	20.0	2,546	5.6	6.6	13.3	15.6	8.9	2,547	16.7	5.2
w/o Topology	20.0	40.0	30.0	2,560	2.1	7.8	12.2	16.7	10.0	2,560	6.2	4.8
w/o Continuity	6.7	13.3	3.3	2,560	2.3	2.6	1.1	4.4	0.0	2,560	7.0	0.4
ACE-off (archived)	20.0	30.0	23.3	2,560	2.6	7.8	13.3	17.8	12.2	2,560	7.7	5.2
TopoPRM (Full)	30.0	43.3	33.3	2,431	5.4	12.3	15.6	18.9	11.1	2,457	7.7	6.4

Algorithm 1 Surface-evidenced support graph construction

Require: Reasoning text y ; optional reference graph $\mathcal{G}^*$
Ensure: Support DAG $\mathcal{G}_y$ and typed evidence ledger

- 1: Extract the reasoning region and segment it into ordered steps $s_1, \dots, s_T$
- 2: **for** $j = 1, \dots, T$ **do**
- 3: Parse normalized expressions, claim keys, variables, and the step type of s_j
- 4: **for** each earlier step $s_i, i < j$ **do**
- 5: Add typed support $i \rightarrow j$ for exact expression or claim reuse
- 6: Add expression-overlap or variable evidence when its guard is satisfied
- 7: **end for**
- 8: **if** s_j is a derivation/conclusion with no structural predecessor **then**
- 9: Add a guarded local-transition edge from s_{j-1}
- 10: **end if**
- 11: **end for**
- 12: Compute orphan support, direction validity, optional reference-edge F1, and continuity
- 13: **return** $\mathcal{G}_y$ and the evidence ledger

Expression, claim, and guarded variable edges receive unit support weight. Local implicit-block and sequential fallback edges contribute 0.5 and 0.3, respectively; incoming support saturates at one. Since construction permits only forward edges, direction and acyclicity diagnose violations of the construction contract rather than semantic validity.

C.2 REPRESENTATION PROPERTIES

Evidence locality and auditability. Every structural contribution resolves to a parsed step, a typed edge with triggering spans, or a continuity decision. This decomposition matters because identical graph-level scores can encode different evidence profiles: exact expression reuse and sequential fallback may produce the same aggregate support while carrying different error risks. The typed ledger exposes that distinction directly, which enables the rule-level false-positive analysis in Section 3.4 without fitting a separate explanation model.

Table 10: Detailed Qwen3.5-9B ablation ledger for CNMO 2024 and MMLU. Metric definitions follow Table 7.

Variant	CNMO 2024 ($n = 30$)						MMLU ($n = 500$)					
	p@1	p@5	maj@5	Tok.	Time	Acc/kTok	p@1	p@5	maj@5	Tok.	Time	Acc/kTok
Base	10.0	16.7	3.3	2,560	1.9	3.9	28.6	41.4	26.2	512	55.1	55.9
Outcome Only	16.7	33.3	26.7	2,530	5.5	6.6	63.0	84.0	61.0	512	146.2	123.0
w/o Topology	20.0	40.0	26.7	2,560	2.0	7.8	65.8	85.0	63.0	512	32.3	128.5
w/o Continuity	6.7	20.0	3.3	2,560	2.4	2.6	20.2	34.8	18.6	512	80.4	39.5
ACE-off (archived)	20.0	30.0	26.7	2,560	2.6	7.8	61.1	83.1	61.2	508	58.1	120.3
TopoPRM (Full)	30.0	47.0	30.0	2,530	2.5	11.9	68.2	86.4	68.8	508	48.0	134.2

Diagnostic non-redundancy. Topology and continuity induce different equivalence classes over traces. Two responses can have the same local grounding rate while differing in whether their conclusions recover non-local premises; conversely, the same supported conclusion can be reached through locally traceable or discontinuous intermediate steps. Keeping both scores explicit preserves these distinctions for reward analysis and defect-conditioned distillation.

Invariances and failure boundary. The representation is invariant to surface changes preserved by normalization, including whitespace, equation delimiters, and several equivalent expression formats. It is not invariant to arbitrary paraphrase, symbol renaming, or mathematically valid step reordering. This asymmetry is intentional: the scorer offers inexpensive, inspectable support evidence rather than semantic equivalence. Continuity can recover local traceability when a non-local edge is missed, while outcome verification remains responsible for mathematically coherent but incorrect traces.

C.3 TOPOLOGY FALSIFICATION TESTS

For this intervention study, q_{dir} and q_{acyc} denote the unaggregated direction and acyclicity checks within q_{topo} . These fixed-trace interventions evaluate the diagnostics, not trained policies. For

Table 11: Matched counterfactual tests of the topology diagnostics on identical source traces. Step reversal primarily changes direction consistency, whereas cycle closure selectively changes acyclicity. The final row quantifies the information lost by averaging the two signals before aggregation.

Intervention	n	Direction response	Acyclicity response
Reverse step order	120	$\Delta q_{\text{dir}} = -0.715$	$\Delta q_{\text{acyc}} = +0.064$
Matched cycle closure	57	$\Delta q_{\text{dir}} = 0.000$	$\Delta q_{\text{acyc}} = -0.328$
Premature averaging	120	$\sigma : 0.110 \rightarrow 0.023$; opposite movement: 39.2%	

reversal, the paired 95% intervals are $[0.657, 0.771]$ for the direction magnitude and $[0.040, 0.091]$ for acyclicity; the two changes correlate at -0.907 ($[-0.976, -0.779]$). Matched cycle closure changes acyclicity by a magnitude in $[0.292, 0.366]$ while leaving direction unchanged. The tests establish selective response to constructed violations, not semantic correctness of the graph.

C.4 AUDIT SAMPLING AND LABELS

The audit samples 120 traces, stratified by task (GSM8K or MATH) and response length (short, medium, or long). The typed rule ledger covers all 120; 119 traces receive both model annotations and supply the 1,770-pair contingency table in Table 3. Each evaluation pack contains parsed steps and candidate pairs but hides the extractor’s decision and edge type. Qwen3-32B and Qwen2.5-32B independently judge whether an earlier step supplies information used later, after which the complete set is manually checked. Precision, recall, and F1 retain the two model annotations and their consensus/union as explicit diagnostic references; raw agreement and Cohen’s κ measure cross-model stability.

Table 12: Representative false-positive edges from the held-out audit. Text is shortened only at sentence boundaries. The examples illustrate three distinct extraction failure modes.

Edge type	Earlier step	Later step	Why the edge is spurious
Variable reference	“For $x = 1$, plug in each integer root to determine a .”	“Thus, the possible values of a are $\{-71, -27, -11, 9\}$.”	A shared symbol is insufficient to identify which case supplies the conclusion.
Order fallback	“The quarters total $10 \times \$0.25 = \2.50 .”	“The dimes total $12 \times \$0.10 = \1.20 .”	Adjacent calculations are parallel branches, not a support relation.
Expression reference	“One hundred patches cost $100 \times \$1.25 = \125 .”	“Selling 100 patches yields $100 \times \$12 = \$1,200$.”	Reusing the count 100 does not make the cost calculation a premise of revenue.

Table 13: Post-hoc rule sensitivity on the 120-trace primary audit. The guarded configuration was designed after inspecting these labels and is evaluated on the same sample; the changes are descriptive and are not attributed to the policies in Table 1.

Evidence type	Released extractor			Guarded extractor			Δ Prec.
	TP	FP	Prec.	TP	FP	Prec.	
Expression reference	49	34	.590	49	34	.590	.000
Expression overlap	19	11	.633	19	11	.633	.000
Implicit block	15	9	.625	26	13	.667	+.042
Order fallback	127	82	.608	196	122	.616	+.009
Variable reference	49	145	.253	23	47	.329	+.076
Overall	259	281	.480	313	227	.580	+.100

Manual verification catches obvious annotation and parsing errors across the full pack, but the reported model-reference metrics are not rebranded as a formal human gold graph. This distinction matters because support can remain ambiguous even after inspection. The release package will include pair-level labels, verification records, aggregate counts, and representative false positives.

C.5 REPRESENTATIVE EXTRACTION FAILURES

Post-hoc rule sensitivity. The typed ledger attributes 145 false positives to variable-reference edges, compared with 34 for expression reference, 11 for expression overlap, 9 for implicit blocks, and 82 for order fallback. This concentration motivates a guarded configuration requiring at least two shared variables for variable-only edges and at least 0.06 token overlap for fallback order edges.

The guarded configuration reduces variable-reference false positives from 145 to 47 and raises overall precision from 0.480 to 0.580 (Table 13). It also changes implicit-block and fallback extraction, making this a multi-rule revision rather than a single-threshold test. Because the same labels informed both design and evaluation, the result is post-hoc and provides no held-out improvement estimate. All policy results use the released extractor.

D ROBUSTNESS AND BOUNDARY TESTS

The following checks separate decoding stability, score association, cross-backbone behavior, and reranking headroom. They constrain the scope of the main results; none substitutes for multi-seed training.

Table 14: Matched DR1-Qwen-7B attribution check on a fixed 200-item GSM8K subset. Greedy columns use one decode; repeated columns summarize five stochastic draws from each fixed policy. Standard deviations measure decoding variation, not training-seed variation.

Reward	Single greedy decode		Five stochastic draws		Structural sources
	p@1	Tokens	p@1 mean $\pm$ sd	Tokens mean $\pm$ sd	
Outcome only	75.5	278.9	68.0 $\pm$ 1.8	198.1 $\pm$ 21.8	none
Outcome + length	76.5	277.5	67.9 $\pm$ 2.7	194.4 $\pm$ 21.7	none
TopoPRM	77.0	438.3	67.8 $\pm$ 1.9	201.3 $\pm$ 25.1	topology + continuity

Table 15: Topology score and correctness on 160 matched GSM8K responses per fixed policy. High and low denote the upper and lower score strata used by the archived analysis.

Policy reward	<i>n</i>	p@1	Mean q_{topo}	$P(\text{correct} \mid \text{high})$	$P(\text{correct} \mid \text{low})$
Outcome only	160	51.25	.925	55.86	40.82
Outcome + length	160	53.12	.928	57.26	41.86
TopoPRM	160	53.12	.939	57.63	40.48

D.1 MATCHED-DECODING STABILITY

With greedy decoding, TopoPRM reaches 77.0 pass@1 at 438 tokens, versus 75.5/279 for outcome-only and 76.5/278 for outcome-plus-length. Across five stochastic draws, the fixed policies obtain 68.0 $\pm$ 1.8, 67.9 $\pm$ 2.7, and 67.8 $\pm$ 1.9 pass@1, with overlapping mean lengths. The greedy ordering is not stable to decoding variation. This check uses a different backbone and harness from Table 6 and says nothing about training-seed variance.

D.2 STRUCTURE-CORRECTNESS ASSOCIATION

The complete reward has the highest mean topology score, 0.939. Its high-score stratum exceeds its low-score stratum by 17.15 accuracy points, but outcome-only exhibits a similar 15.04-point gap. The score is associated with correctness in this sample, yet the comparison does not show unique discrimination attributable to topology training.

D.3 CROSS-BACKBONE CHECK

Table 16: Cross-backbone check on 200 items per benchmark with Llama-3.1-8B-Instruct.

Policy reward	GSM8K		MATH-500	
	p@1	Tokens	p@1	Tokens
Outcome only	85.0	1,008	46.5	3,823
TopoPRM	85.0	619	42.5	3,427

On Llama-3.1-8B-Instruct, TopoPRM preserves GSM8K pass@1 while reducing mean length from 1,008 to 619 tokens. On MATH-500, length falls from 3,823 to 3,427 tokens while pass@1 changes from 46.5 to 42.5. The checkpoint is shorter on both subsets, but does not establish a transferable effect or uniform accuracy preservation.

D.4 FULL BEST-OF-8 LEDGER

Topology-only selection trails the semantic PRM by 3.75 points on GSM8K and 5.00 points on MATH-500. The hybrid ties the PRM on both tasks; both remain below the pool oracle. The fixed fusion therefore yields no aggregate improvement on these pools. Item-level selector disagreements and larger prespecified pools are required to test complementarity.

Table 17: Diagnostic Best-of-8 selection accuracy (%) on one shared Qwen2.5-Math-7B-Instruct candidate pool ($n=80$ per benchmark; 2,048-token generation cap). PRM uses Qwen2.5-Math-PRM-7B; Hybrid multiplies normalized PRM by $1 + q_{\text{topo}}$. Oracle reports whether any candidate is correct. This diagnostic is separate from the primary 4,096-token policy comparison.

Benchmark	Pass@1	Maj@8	PRM@8	Topo@8	Hybrid@8	Oracle@8
GSM8K	93.75	95.00	95.00	91.25	95.00	96.25
MATH-500	83.75	85.00	83.75	78.75	83.75	92.50

E ADDITIONAL ANALYSIS AND INTERPRETATION

How the evidence fits together. Each result targets a different alternative: cross-lineage recurrence varies initialization, the controlled suite fixes it, fixed-trace interventions remove optimization, the blind audit tests support judgments and error localization, and the PRM pool fixes generation. Training seeds, surface references, and pool size leave distinct uncertainties; no result substitutes for the others.

What auditability changes. Scores decompose into step pairs and evidence types, separating expression or claim reuse from variable overlap and fallback order. The audit concentrates error in variable roles while leaving stronger expression matches intact. Guarded rules make the diagnosis actionable but remain post-hoc; attribution requires fresh labels and matched retraining.

Where structural supervision may help. Full and no-topology separate more on OlympiadBench and AIME 2024 than on GSM8K and MATH-500. This motivates, but does not test, greater value from recoverable support when multi-step alternatives remain unresolved. Confirmation should prespecify dependency-depth strata and selector disagreement.

Operating regime. Surface support is most credible with stable symbolic anchors and explicit claims. It weakens under diagrams, paraphrase, or ambiguous symbols: a well-formed graph can encode an invalid deduction, while a correct trace can leave premises implicit. Outcome verification remains the correctness anchor; TopoPRM is more inspectable than a scalar verifier but less complete than formal verification.

F LIMITATIONS AND ARTIFACT CHECKLIST

Upon publication, we will release graph and reward snapshots, training configurations, pair-level audit records, matched diagnostics, and the shared-pool PRM ledger, separated by evidence type.

Graph scoring is training-only, but each controlled configuration has one non-factorial run. Manual verification does not turn model-reference agreement into semantic proof, and guarded rules still require fresh labels and matched retraining. Surface evidence weakens for diagrams, paraphrase, and ambiguous entities. Confirmation requires factorial multi-seed training, a held-out audit, and larger prespecified pools.